

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 03

Factors of Climatic Comfort

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	3.0 - Factors of Climatic Comfort (19 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Method	Question wording and answer-content matched to specific lecture slides. Exact/Direct/Partial describe how much of the requested answer is actually supported by this deck; related A1 topics such as ET/CET, subjective variables and the full thermal-balance equation are not silently credited to Lecture 03.

RESULT

This is a major examination lecture: 24 past-question links were identified across seven of the eight supplied exam years, representing 287 mapped marks. The strongest self-contained block is heat loss/gain in various climatic situations (2018, 2020, 2021, 2022, 2023). Human-body heat exchange is also heavily repeated. Recent vasomotor, thermal-balance and designer-responsibility questions are only partly supported and must be completed from later thermal-comfort material.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 03 establishes the physiological foundation of climatic comfort: four objective climatic factors, body heat production, steady-state body temperature, human heat exchange, and four special climatic heat-loss situations. Historical paper labels vary, so mapping follows the current Section A lecture block rather than the old printed A/B heading.

19	24	7	7	10	287
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Classification	Meaning
Exact	The requested heading/mechanism appears essentially as asked and can be answered directly from the cited slide block.
Direct	Most answer content is present; the student must explain, combine, apply or add examples from the shown material.
Partial	Only part of the requested answer is present; another lecture/class note is required for a complete exam answer.

Source / section note	The deck cover says “Climate & Design (B)”, but it is treated here as Section A - Lecture 03 following the current course organization supplied by the user. Slide 18 also states that designer responsibilities, essential knowledge and complexities belong to the next class; therefore those answers are intentionally marked only Partial here.
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Highest-priority slide blocks

Priority	Slides	Topic	Reason
1	9-15	Heat loss/gain in four climatic situations	Near-identical long questions in 2018, 2020, 2021, 2022 and 2023; this is the single highest-return block.
2	3-8	Thermal comfort, body heat production, steady state and heat exchange	Directly supports 2016, 2018, 2020, 2022, 2023 and 2024 questions; Slide 8 is the core heat-exchange sketch.
3	2	Factors of climatic comfort / objective variables	Short, easy list that directly supports 2016 and 2023 and strengthens broader thermal-comfort answers.
4	12, 17-18	Vasomotor clue + adaptive comfort / designer bridge	Important recent question trend (2022-2024), but the deck does not contain the full autonomic-regulation or designer-responsibility answer.

2. Detailed Year-wise Question Mapping

Question wording is lightly standardized for readability without changing the assessed meaning. "QB p." identifies the page of the uploaded 2016-2024 question bank.

Examinations 2016-2018

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Sec B Q5(b)	20	1	Explain the human-body heat-exchange process.	Direct	3-8	Slides 3-8 establish the heat-generating body, steady-state requirement, heat production and the four heat-exchange paths. Slide 8 is the direct heat-exchange diagram; a 20-mark answer still needs explanation around the diagram.
2016	Sec B Q6(a)	15	1	What are the factors of climatic comfort? Explain with examples.	Direct	2	Slide 2 gives all four requested objective factors exactly; temperature, radiation/absorption, humidity and airflow. The slide does not supply worked examples, so examples must be applied by the student.
2016	Sec B Q6(b)	5	1	What are the thermal-comfort ideas?	Direct	3-5	Slides 3-5 directly frame the idea: the body generates heat, requires a steady state, and may need cooling or warming. The wording is conceptual rather than a formal definition list.
2016	Sec B Q7(b)	5	1	What is the thermal-balance equation?	Partial	3, 7-8	The lecture explains steady state, body temperature and heat-transfer paths, but the actual thermal-balance equation is not printed. Use the later thermal-comfort material/class note for the equation.
2018	Sec A Q4(a)	20	3	Describe the processes of heat loss in various climatic situations.	Exact	9-15	This is the exact major slide sequence: four climatic situations are listed on Slide 9 and explained on Slides 10-15, including the repeated percentage example and dangerous saturated-still-air case.
2018	Sec A Q4(b)	15	3	Explain what may happen in the process of human-body heat exchange, with necessary sketches.	Direct	3-8	Slide 8 directly supplies the four heat-exchange modes and a labelled sketch; Slides 3-7 explain why a steady-state body must release generated heat.
2018	Sec B Q7(a)	20	4	What is thermal comfort? How does the human body produce and release heat to the environment?	Direct	3-8	Slides 3-7 explain thermal-comfort need and body heat production; Slide 8 gives the release mechanisms. The deck does not give a polished textbook definition, so the student must synthesize the opening definition.

2017 diagnostic

No 2017 question was counted as a meaningful Lecture-03 answer-content match. That paper tests Corrected Effective Temperature and other comfort-index material, but those topics belong to Lecture 05 rather than this deck. This prevents the broad "thermal comfort" label from creating a false match.

2. Detailed Year-wise Question Mapping (continued)

Examinations 2020-2022

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2020	Sec A Q1(a)	15	5	Describe human-body heat exchange with the environment through conduction, convection, radiation and evaporation.	Exact	8	Slide 8 is titled "Human Body heat Exchange" and lists exactly the four requested mechanisms, supported by a labelled environmental heat-exchange diagram.
2020	Sec B Q4(b)	15	5	Describe the process of heat loss in various climatic situations.	Exact	9-15	Direct match to the slide-block heading and the four explained situations.
2021	Sec A Q4(a)	13	6	Write the thermal-balance equation of the human body. What happens if the equation is unbalanced?	Partial	3, 7, 12-15	Steady state, body temperature, vasomotor adjustment and dangerous heat accumulation are discussed, but the equation itself is absent. Use later thermal-comfort material for the formal equation and regulation sequence.
2021	Sec B Q5(b)	12	6	Discuss the process of heat loss in various climatic situations.	Exact	9-15	Near-identical to the lecture heading and fully developed across Slides 9-15.
2021	Sec B Q5(c)	10	6	Discuss the designer's responsibilities in assuring thermal comfort in the built environment.	Partial	17-18	Slide 17 gives one explicit design recommendation: aim for a range of comfort with permitted variation. Slide 18 says responsibilities belong to the next class, so this deck is not sufficient for the complete answer.
2022	Sec A Q3(d)	10	8	Describe the human body's mechanism of heat production and write the subjective variables influencing thermal preference.	Partial	6	Slide 6 directly covers heat production (tissue building, energy conversion, muscular movement, food consumption/digestion). Subjective variables are not listed in this lecture.
2022	Sec B Q5(a)	10	8	Discuss the designer's responsibilities in ensuring thermal comfort and mention design tools for moderating microclimate.	Partial	17-18	Only the adaptive-comfort recommendation is present. Formal responsibilities and design tools are explicitly deferred to later material.
2022	Sec B Q5(b)	12	8	Discuss the process of heat loss in various climatic situations.	Exact	9-15	Exact repeated family; all four situations and their mechanisms are in the slide block.
2022	Sec B Q6(d)	6	9	Briefly explain vasomotor adjustment and its response under different ambient temperatures.	Partial	12	Slide 12 mentions vasomotor adjustment as air temperature approaches skin temperature and describes raising skin temperature toward the upper limit. It does not give the full vasodilation/vasoconstriction and autonomic-nervous-system explanation.

2. Detailed Year-wise Question Mapping (continued)

Examinations 2023-2024

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2023	Sec A Q4(b)	8	10	Describe the objective variables by which thermal comfort is affected.	Exact	2	Slide 2 is effectively the requested objective-variable list: temperature, radiation/absorption, humidity and airflow.
2023	Sec A Q4(c)	10	10	Write the significance of thermal comfort for the human body.	Direct	3-7	Slides 3-7 explain why comfort matters: the body is a heat-generating machine that requires a steady state and maintenance of deep-body temperature.
2023	Sec A Q4(d)	12	10	Write the human thermal-balance equation and describe regulatory mechanisms when it is unbalanced.	Partial	3, 7, 12-15	The deck gives steady-state logic, vasomotor reference and heat-accumulation consequences, but not the equation or a complete hot/cold regulatory flow.
2023	Sec B Q5(a)	8	11	Explain vasomotor adjustment under different ambient/climatic conditions.	Partial	12	Only the hot-condition vasomotor clue is provided. Full vasodilation/vasoconstriction behavior under both hot and cold conditions requires later material.
2023	Sec B Q6(c)	14	11	Discuss the process of heat loss or gain in various climatic conditions.	Exact	9-15	This is the repeated core block. Slides 11-15 also show when convection/radiation cease to cool and when the body may gain heat or store heat.
2024	Sec A Q3(c)	12	13	What is thermal comfort? How does the human body maintain thermal balance?	Direct	3-8, 12-15	The lecture explains steady-state need, heat production, normal deep-body temperature, heat-exchange paths and consequences of inadequate heat loss. It does not print the formal balance equation, but the question does not explicitly require derivation.
2024	Sec B Q6(a)	10	14	Explain vasomotor adjustment as part of human thermal regulation and relate temperature variation/autonomic response to comfortable indoor design.	Partial	12, 17	Slide 12 gives only a partial vasomotor mechanism; Slide 17 provides the adaptive-design idea of permitting a range of comfort. The full autonomic-regulation and design analysis must come from later notes.
2024	Sec B Q6(b)	10	14	Discuss the designer's responsibilities in thermal comfort and the essential-knowledge domains required.	Partial	17-18	Slide 17 provides one general comfort-design principle; Slide 18 explicitly announces responsibilities and essential knowledge as the next class, confirming the gap.

Recent-paper signal

The 2023-2024 papers broaden the lecture from basic heat exchange into objective variables, vasomotor regulation, body balance and designer interpretation. Use Lecture 03 as the physiological base, but do not treat it as a substitute for the later equation, subjective-variable or designer-responsibility material.

3. Repeated Question Patterns

The patterns below are ordered by exam return. A repeated cluster means the tested mechanism or decision is substantially the same, even when wording or section labels change.

#	Pattern	Years / questions	Slides	Coverage / exam insight
1	Heat loss/gain in various climatic situations	2018 Q4(a); 2020 Q4(b); 2021 Q5(b); 2022 Q5(b); 2023 Q6(c)	9-15	5 appearances; strongest exact block in the lecture. Learn the four situations in sequence, not as isolated points.
2	Human body heat exchange: conduction, convection, radiation, evaporation	2016 Q5(b); 2018 Q4(b); 2020 Q1(a); also supports 2018 Q7(a), 2024 Q3(c)	3-8	The heat-exchange sketch is compulsory. Slide 8 is the core, Slides 3-7 supply the physiological reason.
3	Thermal comfort / heat production / steady state	2016 Q6(b); 2018 Q7(a); 2022 Q3(d); 2023 Q4(c); 2024 Q3(c)	3-7	Repeated conceptual family. The deck is strong on “why” comfort matters and how heat is produced.
4	Factors of climatic comfort / objective variables	2016 Q6(a); 2023 Q4(b)	2	Simple high-value list: temperature, radiation/absorption, humidity, airflow.
5	Thermal balance equation / imbalance regulation	2016 Q7(b); 2021 Q4(a); 2023 Q4(d); related 2024 Q3(c)	3, 7, 12-15	Repeated but only partial here: the equation is missing. Do not invent it from this slide deck.
6	Vasomotor adjustment / autonomic regulation	2022 Q6(d); 2023 Q5(a); 2024 Q6(a)	12	Recent 3-year pattern. Lecture 03 contains only one hot-condition clue, so full preparation must continue elsewhere.
7	Designer responsibility for thermal comfort	2021 Q5(c); 2022 Q5(a); 2024 Q6(b)	17-18	Slide 17 gives adaptive variation; Slide 18 explicitly defers full responsibilities/knowledge to the next class.

Quick recognition rules

Question clue	Go first to	What Lecture 03 can do
“heat loss/gain in various climatic situations”	Slides 9-15	Exact block. Write all four cases and explain how convection, radiation and evaporation change.
“conduction, convection, radiation, evaporation”	Slide 8 (+ 3-7)	Direct heat-exchange sketch and mechanisms.
“body heat production / metabolism”	Slide 6	Direct list of exothermal activities and food/digestion energy source.
“factors of climatic comfort / objective variables”	Slide 2	Direct four-factor list.
“steady state / deep body temperature”	Slides 3, 7	Physiological base for thermal comfort and balance discussions.
“vasomotor adjustment”	Slide 12, then later thermal-comfort material	Only partial here; do not claim a full hot/cold autonomic sequence from this deck.
“thermal balance equation”	Use another source after Slides 3, 7-8	Lecture 03 has the idea of steady state but not the equation.
“designer responsibilities / essential knowledge”	Slide 17 for one principle; Lecture 04 for full answer	Slide 18 explicitly says the next class covers these topics.
“subjective variables / ET / CET / EW / OT”	Do not use Lecture 03	These are outside this deck; use the later comfort-index/thermal-comfort materials.

4. Slide-to-Question Reverse Index

Study slide blocks, then immediately practice the linked past questions. "Partial" means the slide is useful evidence but not a complete answer source.

Slides	Topic	Past questions	Study note
2	Four factors of climatic comfort	2016 Q6(a); 2023 Q4(b)	Very high return per slide; memorize the exact four-factor list.
3-5	Why thermal comfort? heat-generating body; cooling/warming; steady-state idea	2016 Q6(b); 2018 Q7(a); 2023 Q4(c); 2024 Q3(c)	Conceptual foundation. Use these slides to open long answers before mechanisms.
6	Human body heat production	2018 Q7(a); 2022 Q3(d); 2024 Q3(c)	Direct support for heat-production questions; subjective variables are not here.
7	Human-body steady state / deep-body temperature	2016 Q7(b); 2021 Q4(a); 2023 Q4(c,d); 2024 Q3(c)	Useful physiological support, but not the thermal-balance equation.
8	Human body heat exchange: four mechanisms + sketch	2016 Q5(b); 2018 Q4(b), Q7(a); 2020 Q1(a); 2024 Q3(c)	One of the highest-value single slides. Redraw the labelled sketch in the exam.
9-10	Four situations list + calm/warm moderate-RH case	2018 Q4(a); 2020 Q4(b); 2021 Q5(b); 2022 Q5(b); 2023 Q6(c)	Start the repeated long answer here; remember the slide example 45% radiation, 30% convection, 25% evaporation.
11-12	Hot air + considerable radiation; skin-temperature / vasomotor clue	Same heat-loss family; 2022 Q6(d); 2023 Q5(a); 2024 Q6(a)	Excellent for explaining why convection falls as air temperature approaches skin temperature; vasomotor detail remains incomplete.
13-14	Hot air + radiation + air movement; evaporative boundary-layer effect	Same heat-loss family	Critical mechanism: moving air replaces saturated air at the skin and accelerates evaporation while RH remains below saturation.
15-16	Saturated still air above skin/body temperature; heat-stroke illustration	Same heat-loss family; balance questions as support	Use as the dangerous limiting case. Specific medical threshold numbers are not directly required by the QB, so prioritize mechanism over memorizing extreme values.
17	Effects of prolonged exposure / recommendation for variation	2021 Q5(c); 2022 Q5(a); 2024 Q6(a,b)	One valuable design principle: comfort should allow variation rather than remain perfectly fixed. Not a full designer-responsibility list.
18	Next class: responsibilities, essential knowledge, complexities	2021 Q5(c); 2022 Q5(a); 2024 Q6(b)	A gap marker, not answer content. It tells you to continue to Lecture 04.
19	Reference / thank you	No direct question	Navigation only.

5. Coverage Gaps and Diagnostic Exclusions

This page prevents a common exam-preparation error: treating every A1 thermal-comfort question as if Lecture 03 alone can answer it.

Gap	Affected questions	Action
Full human thermal-balance equation and complete imbalance-regulation sequence	2016 Q7(b); 2021 Q4(a); 2023 Q4(d); related 2024 Q3(c)	Lecture 03 explains steady state and heat loss, but does not print the equation. Use later thermal-comfort notes/class material.
Subjective variables affecting thermal preference	2018 Q7(b); 2022 Q3(d); 2024 Q3(d)	Not listed in Lecture 03. Do not infer them from the body-temperature slides.
Full vasomotor adjustment: vasodilation, vasoconstriction and autonomic nervous system under hot/cold conditions	2022 Q6(d); 2023 Q5(a); 2024 Q6(a)	Slide 12 only mentions vasomotor adjustment in the hot-air case. Complete this from later thermal-comfort material/class notes.
Designer responsibilities, essential knowledge and climate-design complexities	2021 Q5(c); 2022 Q5(a); 2024 Q6(b)	Slide 17 gives one recommendation; Slide 18 explicitly says these are in the next class. Use Lecture 04.
ET, CET, EW, OT, effective-temperature limitations and bioclimatic chart	2016 Q6(c); 2017 Q7(a-v); 2018 Q7(c); 2021 Q6(c); 2022 Q6(c); 2023 Q5(b)	These belong to Lecture 05 / comfort-index material, not Lecture 03.
Passive vs active thermal control	2018 Q8(a); 2020 Q4(a)	Lecture 03 discusses the need to cool/warm the body but does not classify passive/active building control.
Precise medical thresholds on Slides 7 and 15	No direct QB question asks for the full threshold list	The source presents overlapping/strong threshold statements. For this course mapping, prioritize the steady-state and heat-stress mechanism rather than treating these numbers as a medical reference.

Mapping discipline	A1 appears in all eight papers as a broad set, but Lecture 03 itself does not cover every A1 subtopic. The 2017 paper is the clearest example: it tests comfort indices, so it is intentionally excluded from this lecture-specific link count. The same rule is used for subjective variables and passive/active control.
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6. Recommended Study Plan

Study by repeated exam pattern rather than slide order. The long-answer heat-loss block comes first because it has five near-identical appearances.

Step	Study block	Slides	Practice immediately
1	Four heat-loss/gain situations - learn sequence + mechanism	9-15	2018 Q4(a); 2020 Q4(b); 2021 Q5(b); 2022 Q5(b); 2023 Q6(c)
2	Human-body heat exchange + labelled sketch	8 (+ 3-7)	2016 Q5(b); 2018 Q4(b); 2020 Q1(a)
3	Thermal comfort, heat production and steady state	3-7	2016 Q6(b); 2018 Q7(a); 2022 Q3(d); 2023 Q4(c); 2024 Q3(c)
4	Four factors / objective variables	2	2016 Q6(a); 2023 Q4(b)
5	Vasomotor clue in hot air	12	2022 Q6(d); 2023 Q5(a); 2024 Q6(a) - then complete from later material
6	Adaptive comfort / prolonged-exposure design recommendation	17	2021 Q5(c); 2022 Q5(a); 2024 Q6(a,b) as partial support
7	Gap handoff to next class	18	Move to Lecture 04 for responsibilities, essential knowledge and complexities

Exam-ready minimum from Lecture 03

- Slide 2: memorize Temperature - Radiation/Absorption - Humidity - Airflow.
- Slide 8: redraw the four-path human heat-exchange diagram and explain conduction, convection, radiation and evaporation.
- Slides 9-15: answer the four climatic situations in order; understand why sensible heat loss falls near skin temperature and why moving air helps evaporation.
- Slide 6: know tissue building, energy conversion and muscular movement as heat-producing/exothermal activities, with food/digestion as the energy source.
- Slides 3 and 7: use the steady-state idea to explain why the body must balance produced/gained heat with released heat.
- Slide 12: remember it as a vasomotor clue only - not a complete hot/cold autonomic-regulation answer.
- Slide 17: one design principle - aim for a range of comfort conditions with variation; prolonged unchanging conditions can themselves become uncomfortable.
- Slide 18: treat it as a handoff marker to Lecture 04, not as answer content.

Final recommendation

If time is short, prioritize Slides 9-15 first, then Slide 8, then Slides 3-7 and Slide 2. Those blocks contain every Exact match and most Direct matches. Use Slides 12 and 17 only as supporting evidence, and explicitly move to later material for the full thermal-balance equation, vasomotor regulation and designer-responsibility answers.